

4. A beam of monochromatic light is shone normally (at right angles) on to a diffraction grating.

(a) **Explain** in clear steps why bright beams emerge from the grating at angles,  $\theta$ , to the normal given by the equation:

$$d \sin \theta = n\lambda$$

[3]

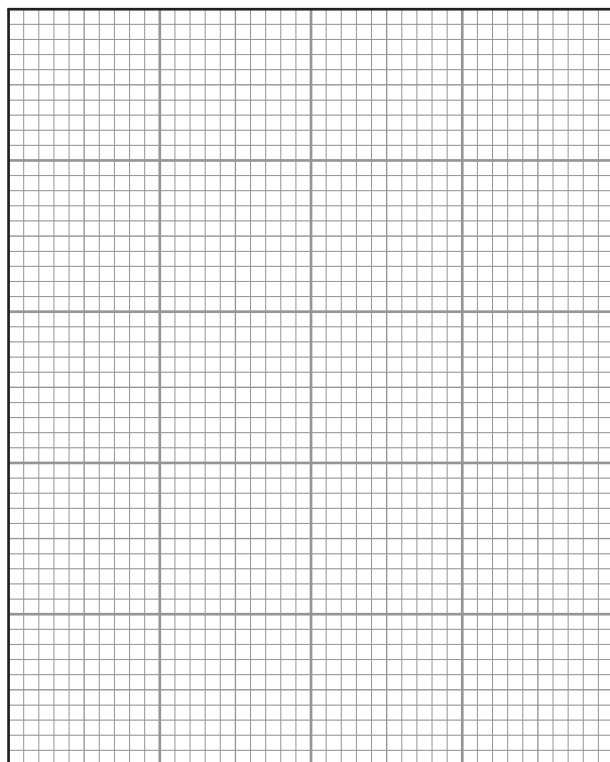
(b) The angles,  $\theta$ , at which the bright beams emerge are given in the table below.

| Order, $n$ | $\theta$ (mean) |  |
|------------|-----------------|--|
| 0          | 0               |  |
| 1          | 16              |  |
| 2          | 35              |  |
| 3          | 58              |  |

(i) Plot a graph of  $\sin \theta$  ( $y$ -axis) against  $n$  ( $x$ -axis) on the grid provided.

[3]





- (ii) **Use your graph** to determine the wavelength of the light. The separation between the centres of slits in the grating is 1800 nm. Show your working. [3]

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